CASE 17
AUTOIMMUNE POLYENDOCRINOPATHY–CANDIDIASIS–ECTODERMAL DYSTROPHY (APECED)

Role of the thymus in the negative selection of T lymphocytes.

Millions of different T-cell antigen receptors are generated during the development of thymocytes in the thymus gland. This stochastic process inevitably leads to the formation of some T-cell antigen receptors that can bind to self antigens. When a T-cell receptor of a developing thymocyte is ligated by antigen on stromal cells in the thymus, the thymocyte dies by apoptosis. This response of immature T lymphocytes to stimulation by antigen is the basis of negative selection (Fig. 17.1). Elimination of these T cells in the thymus aborts their subsequent potentially harmful activation.

Experiments in mice have shown that self-MHC:self-peptide complexes encountered in the thymus purge the T-cell repertoire of immature T cells bearing self-reactive receptors. The bone marrow-derived dendritic cells and macrophages of the thymus are thought to have the most significant role in negative selection (Fig. 17.2), but thymic epithelial cells may also be involved.

Most of what we know about negative selection has come from studying transgenic mice. How the myriad of self antigens, many of them organ specific, can be 'seen' in the developing thymus gland by developing thymocytes remained largely undefined until the genetic defect in the human autosomal recessive disease APECED (autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy) was identified. The gene that is mutated in this disease is called *AIRE* (for *a*uto*i*mmune *re*gulator), and its discovery has given new insight into the process of negative selection of thymocytes.

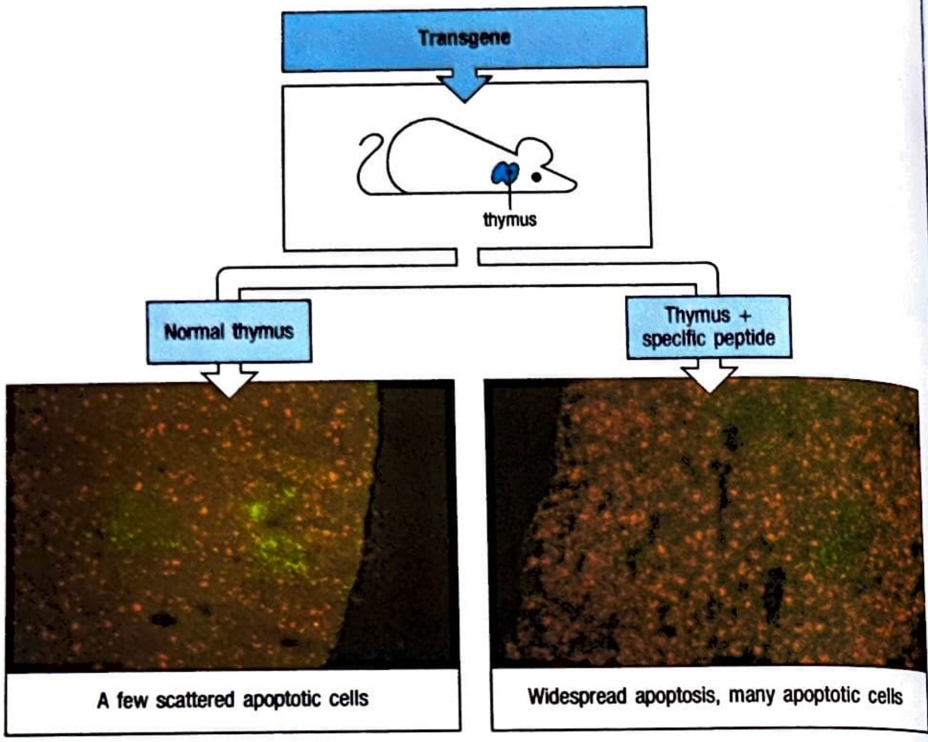
TOPICS BEARING ON THIS CASE:

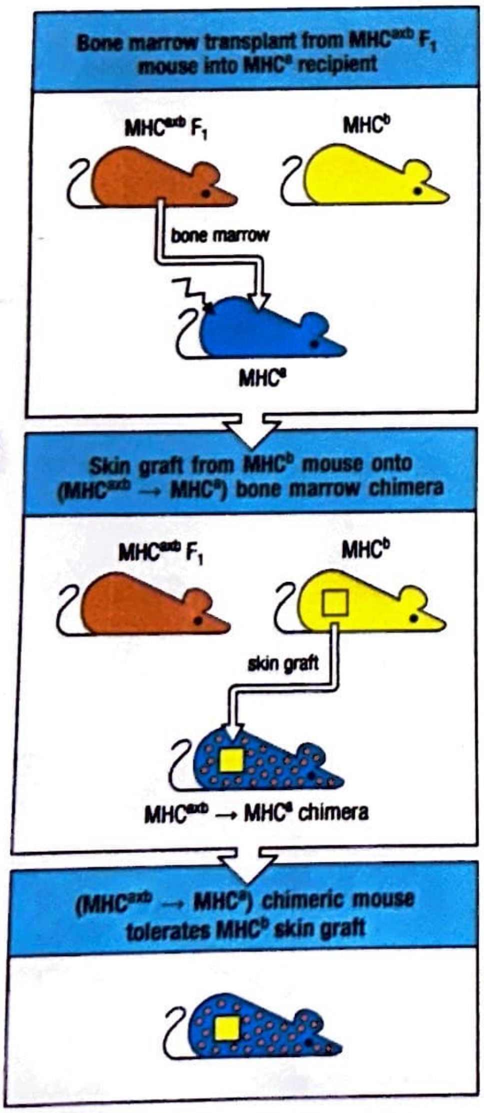
Lymphocyte development

Negative selection of T lymphocytes

Autoimmune disease

Fig.17.1 T cells specific for self antigens are deleted in the thymus. In mice transgenic for a T-cell receptor that recognizes a known peptide antigen complexed with self MHC, all the T cells have the same specificity. In the absence of the peptide, most thymocytes mature and emigrate to the periphery. This can be seen in the lower left panel, in which a normal thymus has been stained with antibody to identify the medulla (in green), and by the TUNEL technique to identify apoptotic cells (in red). If the mice are injected with the peptide that is recognized by the transgenic T-cell receptor, then massive cell death occurs in the thymus, as shown by the increased numbers of apoptotic cells in the lower right-hand panel. Photographs courtesy of A. Wack and D. Kioussis.





The case of Robert Jordan: a young man with a lifelong history of autoimmune diseases of various endocrine glands.

When Robert was 18 months old his mother took him to the pediatrician, complaining that his skin was very dry and that his movements seemed sluggish compared with those of a normal active infant. Robert's pediatrician made the diagnosis of hypothyroidism. He prescribed Synthroid (thyroid hormone) and in a few weeks the child was much improved. But by the time Robert was almost 6 years old his mother felt that he was not growing at a normal rate. The pediatrician referred Robert to Dr Hemingway, an endocrinologist at the Children's Hospital. On examination he found that Robert's height and weight were below the third centile for his age. An X-ray of Robert's wrist revealed a bone age of 3 years 6 months when he was, in fact, 5 years 10 months old.

A blood test showed that Robert's level of thyroid-stimulating hormone (TSH) was more than 60 ng dl^{-1}. This is several times the upper limit of normal TSH levels and indicated that Robert was receiving inadequate thyroid hormone replacement. The dose of Synthroid was increased from 0.05 to 0.075 mg per day. Robert resumed normal growth and his bone age subsequently caught up with his actual age.

Fig. 17.2 Bone marrow-derived cells mediate negative selection in the thymus. When $MHC^{a \times b}$ F1 bone marrow is injected into an irradiated MHC^a mouse, the T cells mature on thymic epithelium expressing only MHC^a molecules. Nevertheless, the chimeric mice are tolerant to skin grafts expressing MHC^b molecules (provided that these grafts do not present skin-specific peptides that differ between strains a and b). This implies that the T cells whose receptors recognize self antigens presented by MHC^b have been eliminated in the thymus. As the transplanted $MHC^{a \times b}$ F_1 bone marrow cells are the only source of MHC^b molecules in the thymus, bone marrow-derived cells must be able to induce negative selection.

As Dr Hemingway pondered the cause of Robert's hypothyroidism, he learned from Mrs Jordan that she had another child, a daughter 2 years older than Robert. This girl had nail dystrophy, perioral candidiasis, hypoparathyroidism, and serum antibodies against islet cells of the pancreas (but no clinically apparent diabetes). This led Dr Hemingway to suspect that Robert had the inherited disease APECED, an autosomal recessively inherited abnormality. He tested Robert's serum calcium levels to make sure that he had not developed hypoparathyroidism like his sister; they were normal. Neither did Robert's serum contain antibodies against any other endocrine glands.

Fig. 17.3 Abnormal growth (dysplasia) of fingernails in APECED. Photograph courtesy of Mark S. Anderson.

Dr Hemingway had noticed that Robert's fingernails were thickened, with longitudinal notches and ridging (Fig. 17.3). He referred Robert to a dermatologist, who agreed that this abnormality was consistent with APECED. Nail scrapings cultured for *Candida albicans* were negative. The dermatologist also noticed two patches of hair loss at the top and back of Robert's scalp. He could easily pull out the hair, whose roots looked atrophied under the microscope, and made the diagnosis of alopecia areata (patchy hair loss).

Age 6, growth less than normal. Increase thyroid hormone.

Robert continued to grow satisfactorily, and at 8 years old his TSH level was 2.5 ng dl^{-1}, a normal level which indicated adequate thyroid hormone replacement. But he continued to lose hair in patches and lost his eyebrows. He developed fissures at the angle of his mouth as a result of infection with *C. albicans*. He was taunted at school about his bizarre appearance, and his schoolwork deteriorated. With his parents' divorce pending, Robert became depressed and took an overdose of Synthroid in a suicide attempt. He received intensive psychotherapy for his depression.

18-month-old boy with lethargic movements and dry skin.

As Robert approached puberty, his scrotum became darkly pigmented, as did the areolae around his nipples. Dr Hemingway suspected that Robert was developing adrenal insufficiency (Addison's disease). A blood test revealed a level of adrenocorticotropic hormone (ACTH) three times the upper limit of normal. The doctor prescribed the steroid prednisone at 5 mg per day and Fluorinef (which conserves sodium and potassium excretion) at 0.1 mg a day.

Older sibling with various autoimmune conditions. Inherited autoimmune disease? APECED?

When he was 18 years old Robert noticed that he had started to bruise easily and that his gums bled after brushing his teeth. Dr Hemingway sent him to a hematologist, who found a low platelet count of 34,000 μl^{-1}. The hematologist decided that Robert had developed yet another autoimmune disease, idiopathic thrombocytopenic purpura (ITP).

APECED confirmed in the patient and sister by genetic testing.

Autoimmune polyendocrinopathy–candidiasis–ectodermal dystrophy (APECED).

Age 16, Addison's disease and idiopathic thrombocytopenic purpura have developed.

APECED is also known as autoimmune polyglandular syndrome (APS) type I. Although there are several polyglandular diseases of unknown origin, APECED is the only one known to have a pattern of autosomal recessive inheritance. Affected individuals such as Robert Jordan develop a wide range of autoantibodies not only against organ-specific antigens of the endocrine glands but also against antigens in the liver and skin, and, as we have seen in this case, against blood cells such as platelets. In addition to these autoimmune diseases, affected individuals have abnormalities of various ectodermal elements such as fingernails, teeth, and skin. They also have an increased susceptibility to infection with the yeast *Candida albicans*. Interestingly, this increased susceptibility to candidiasis has an autoimmune basis. In particular, patients with APECED produce autoantibodies to interleukin-17A (IL-17A), IL-17F, and IL-22, which play a critical role in the defense against *Candida* species (see Case 20). Finally, patients with APECED also produce antibodies to interferon-α (IFN-α) and to IFN-ω.

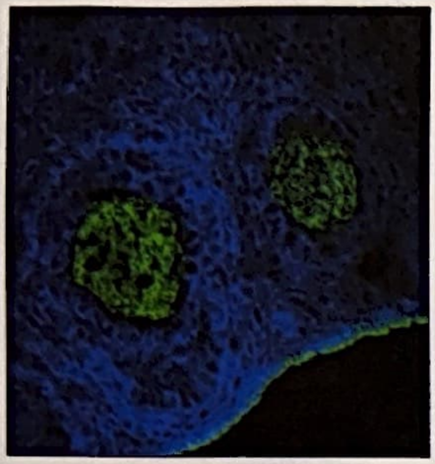


Fig. 17.4 Ovarian follicles stained with serum from a patient with APECED. Serum was reacted with a microscopic section of ovary and then with fluorescent antibody against human IgG, which reveals serum IgG bound to antigens in the zona pellucida around the oocyte (green). Photograph courtesy of Mark S. Anderson.

APECED has a high incidence among Finns, Sardinians, and Iranian Jews, as high as 1 in 5000 to 1 in 9000 births in some populations. This made it possible to map the gene responsible to chromosome 21 in affected Finnish families. The gene was cloned and named *AIRE* for autoimmune regulator. It was the first identified gene outside the major histocompatibility complex to be associated with autoimmune disease, and seems to encode a transcriptional regulator. Robert and his sister were found to have a deletion of 13 base pairs in exon 8 of the gene. Their parents were both heterozygous for this mutation.

When the equivalent gene *Aire* was 'knocked out' in mice by homologous recombination, they developed autoimmune diseases just like the human patients; the extent and severity of these autoimmune diseases progressed as the mice aged. The protein AIRE was found normally to be expressed predominantly in the epithelial cells of the medulla of the thymus and only weakly in peripheral lymphoid tissue. AIRE induces the expression of 200–1200 genes in the thymic medulla, including genes that encode organ-specific antigens of the salivary glands and the zona pellucida of ova (Fig. 17.4), among others. In normal circumstances, the expression of these self peripheral tissue antigens in the thymus permits negative selection of those developing T cells that react to them. When AIRE is lacking, these antigens are also not present. The potentially self-reactive T cells are not removed from the repertoire in the thymus and leave to cause autoimmune reactions in peripheral organs. The molecular mechanisms by which AIRE controls the expression of peripheral tissue antigens in the thymic medulla are complex, and involve recognition of unmethylated histone 3, transcriptional activation, and control of mRNA splicing.

Questions.

1 *What caused the deep pigmentation of Robert's scrotum and the areolae around his nipples that led to the suspicion that he had developed an autoantibody against his adrenal cortical cells (Addison's disease)?*

2 *What abnormality might be found in the lymph nodes of the mice that lacked the Aire gene?*

3 *When lymphocytes from Aire-deficient mice were transferred into Rag-deficient mice, which have no mature lymphocytes of their own, autoimmune disease developed in the recipient. This did not occur when lymphocytes from normal mice of the same strain were transferred into Rag-deficient mice. What can be deduced from this experiment?*

4 *Primary ovarian failure has frequently been observed in females with APECED. What is the underlying mechanism?*